Temperature survey

I1a
Core

Aim

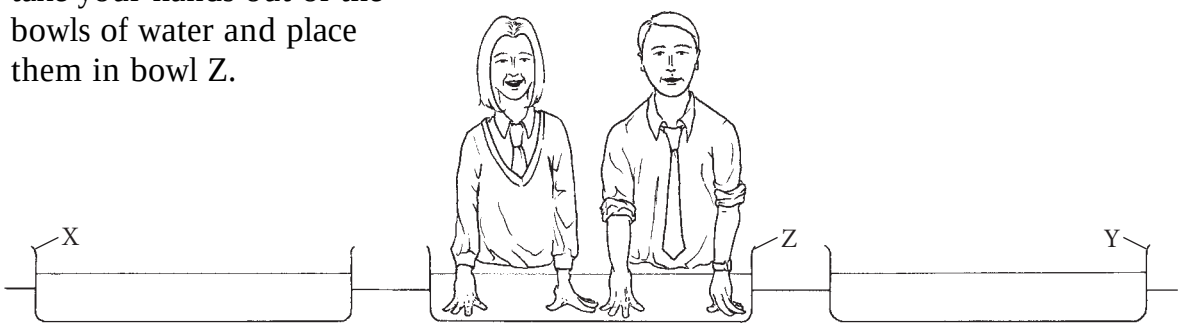
To compare temperatures of water by feeling with your hand and by measuring with a thermometer.

Equipment

- bowls of water at different temperatures
- thermometer
- paper towels

What to do

1. Make a table like the one below ready to record your results.
2. Work in pairs. Decide who will start with bowl X (student X) and who will start with bowl Y (student Y).
3. Student X places their hands in bowl X. Student Y places their hands in bowl Y.
4. At the same time, both take your hands out of the bowls of water and place them in bowl Z.



5. Dry your hands. Together write down in both your tables a description of how warm or cold the water felt to you in bowls X and Y and then how it felt in bowl Z.
6. Using a thermometer, measure the temperature of the water in each bowl.
7. Record the temperature readings in your table.

Results

	Bowl X	Bowl Y	Bowl Z
Description of water temperature (student X)			
Description of water temperature (student Y)			
Temperature measured using the thermometer			

Questions

1 Is your sense of touch a good, reliable way of measuring temperature? Use the results in your table to explain your answer.

2 Why is it better to use numbers such as 20 °C than to use words such as hot, cold, warm?

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Looking at different types of thermometer

Aim

To compare types of thermometer.

Equipment

- mercury-in-glass stirring thermometer (−10 to 110 °C) in a beaker of ice/water
- mercury-in-glass long thermometer (−10 to 110 °C) in a beaker of warm water
- mercury-in-glass thermometer (−10 to 300 °C) in a beaker of hot water
- clinical thermometer
- alcohol thermometer
- temperature sensor and data logger in a beaker of warm water
- temperature-dependent plastic

What to do

You are going to look at a number of different thermometers that are recording the temperatures of different materials.

For each thermometer:

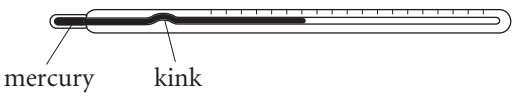
❶ Record the temperature reading.

❷ Sketch and briefly describe the thermometer.

Questions

1 Which thermometer would you choose for each of the following temperature measurements?
- **a** body temperature
- **b** hot water
- **c** the temperature inside a refrigerator
- **d** the exact boiling point of a liquid thought to have a boiling point between 150 and 250 °C.

2 A clinical thermometer has a kink in the capillary tube, as shown in the diagram.



Explain the purpose of the kink.

3 If you were on an expedition to the South Pole, you would not take a mercury-in-glass thermometer with you.
- **a** Why not?
- **b** Which type of thermometer would you use? Give a reason for your answer.

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Warming up and cooling down

Aim

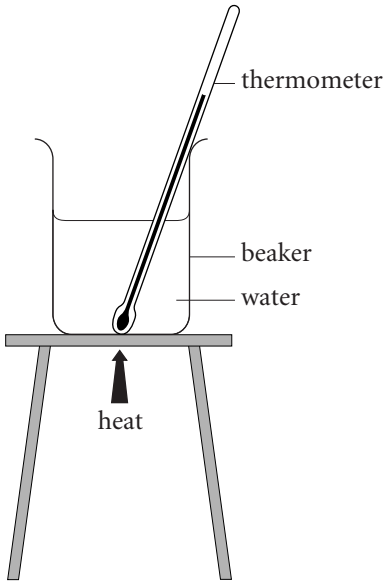
To investigate heating and cooling.

Equipment

- beaker
- Bunsen burner
- graph paper
- tripod and gauze
- thermometer
- stopwatch
- heatproof mat

What to do

1. Make a table like the one below to record your results.
2. Pour about 150 cm^3 of water into the beaker.
3. Place the beaker on the tripod and gauze.
4. Record the temperature of the water in row A at time 0 minutes.
5. Start the stopwatch. Heat the water and record the temperature once every minute until it has been boiling for 3 minutes.
6. Turn off the Bunsen burner and reset the stopwatch.
7. In row B, record the temperature of the water, once every minute, as it cools for about 10 minutes.
8. Draw a graph of temperature (*y*-axis) against time (*x*-axis) to show the heating of the water, using the readings in row A.
9. Draw a graph of temperature (*y*-axis) against time (*x*-axis) to show the cooling of the water, using the readings in row B.



Results

Time (minutes)	0	1	2	3	4	5	6	7	8	9	10
A Temperature during heating (°C)											
B Temperature during cooling (°C)											

Questions

1 a From the evidence on your graphs, which gives the faster temperature change, the heating or cooling of the water?

b Why do you think there is a difference between the rate of heating and the rate of cooling?

2 Jenny says the energy in the hot water disappears as the water cools. Is she right? Explain your answer.

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Conduction in metals

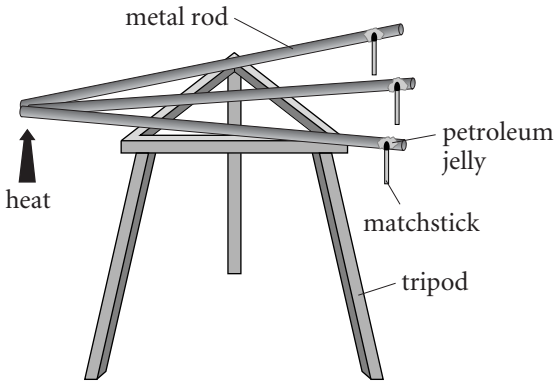
Aim

To investigate thermal conduction in metals.

Equipment

- three metal rods – brass, copper and iron
- tripod
- heatproof mat
- Bunsen burner
- petroleum jelly
- three spent matchsticks

What to do



1. Identify the three metal rods. (Hint: Copper has a reddish colour and brass is more yellow.)
2. Place the rods on the tripod (without a gauze) so that they are fanned out as shown in the diagram.
3. Stick a matchstick to each rod, using a small amount of petroleum jelly. The matchsticks should all be the same distance from the ends of the rods to be heated, as in the diagram.
4. Heat the rods at the end where they touch each other.
5. Observe the rods and note down the order in which the matchsticks fall off.

Questions

1 Which metal is the best thermal conductor?

2 Why was it important that all the matchsticks were the same distance from the ends of the rods being heated?

3 Can you suggest any ways to make this a more accurate experiment?

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Conduction in water

Aim

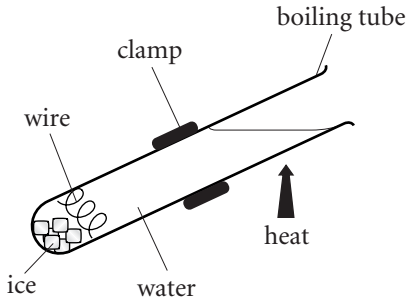
To show that water is a poor conductor of heat.

Equipment

- boiling tube
- ice
- coil of wire
- Bunsen burner
- heatproof mat
- clamp stand
- glass rod



What to do



1. Put some ice in the bottom of the boiling tube.
2. Using the glass rod, push the coil of wire down the tube so that it will hold the ice at the bottom when the tube is filled with water.
3. Fill the tube to about 1 cm from the top with water.
4. Clamp the tube so that it is arranged as shown in the diagram.
5. Heat the water at the **top** of the tube. Observe the ice carefully.
6. When the water at the top of the tube starts to boil, stop heating.

Questions

1 What did you notice about the ice when the water at the top of the tube was boiling?

2 Is water a good conductor of heat? Explain how this experiment shows this.

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Insulators

Aim

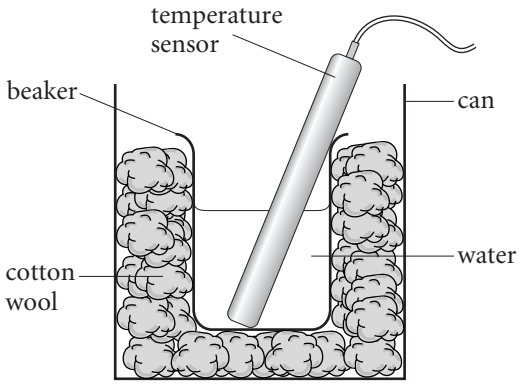
To investigate a variety of insulators.

Equipment

- beaker
- polystyrene cup
- cotton wool
- can
- supply of hot water
- stopwatch
- temperature sensor and data logger

What to do

1. Pour about 200 cm^3 of the hot water into the beaker.
2. Start the stopwatch and take readings of the temperature using the temperature sensor and data logger.
3. Repeat the experiment using the polystyrene cup instead of the beaker.
4. Repeat the experiment again, putting the beaker in the can and surrounding it with cotton wool as shown in the diagram.



Questions

1 Which material is the best insulator? Give examples from your readings to justify your conclusion.

2 Suggest some ways in which you could improve the experiment.

3 What is it about the structure of the best insulator that makes it a good insulator?

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Insulators

Aim

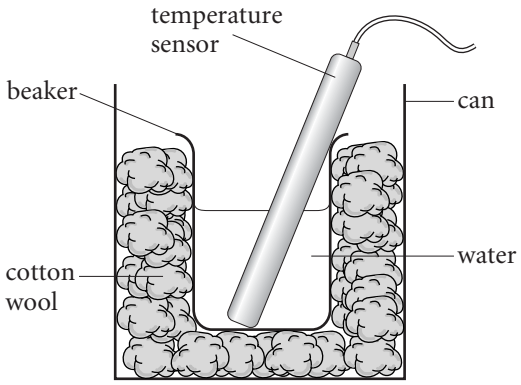
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Expansion and contraction in solids

Aim

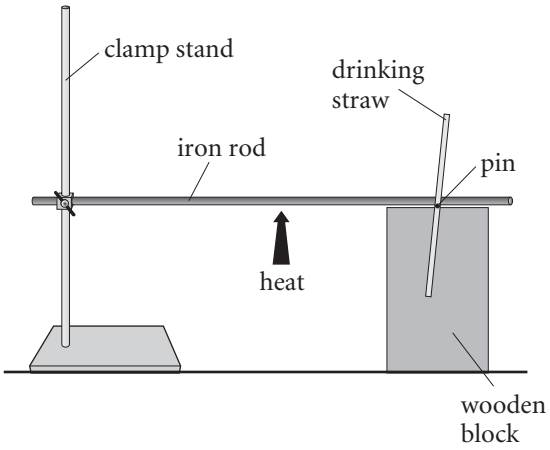
To demonstrate the expansion and contraction of a solid.

Equipment

- iron rod
- clamp stand
- pin
- drinking straw
- Bunsen burner
- heatproof mat
- block of wood

What to do

1 Watch as your teacher demonstrates the experiment, then answer the questions.



Questions

1 When the rod was heated, which way did the straw rotate (clockwise or anticlockwise)?

2 What happened to the rod when it was heated?

3 Explain how the pin was used to make a small movement into a large movement that could easily be seen.

4 Give three examples of instruments that use a similar idea to the pin and straw to make a small movement easily seen.

5 **a** What happened to the straw when the rod was allowed to cool?

b What does this tell you about what happens to the rod as it cools?

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Expansion and contraction in liquids

Aim

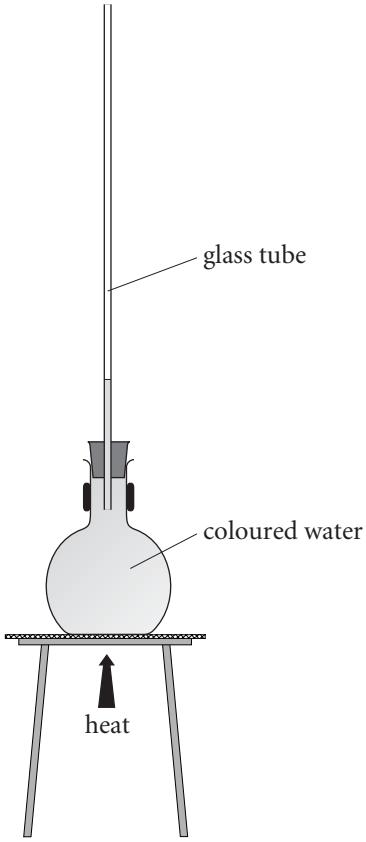
To demonstrate the expansion and contraction of a liquid.

Equipment

- large flat-bottomed flask filled with coloured water, with a bung and a long glass tube
- tripod and gauze
- heatproof mat
- Bunsen burner
- clamp stand

What to do

1. Watch as your teacher demonstrates the experiment, then answer the questions.



Questions

1 When the water was heated, what happened to the water level in the tube?

2 Why did this happen?

3 What happened to the water level in the tube when the water was allowed to cool?

4 Why did this happen?

5 Which instrument works on the same principle as this experiment?

6 Use the particle model to write a brief explanation of the expansion of water when it is heated.

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Expansion and contraction in gases

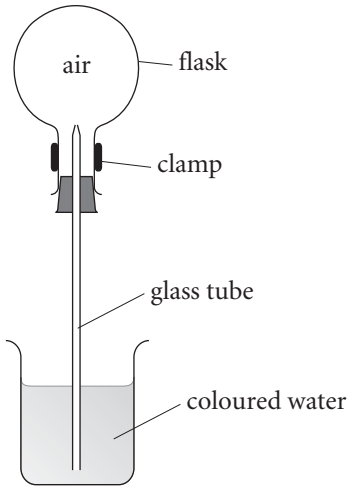
Aim

To investigate the expansion and contraction of a gas.

Equipment

- round-bottomed flask with a bung and a glass tube
- clamp stand
- beaker filled with coloured water
- Bunsen burner
- heatproof mat

What to do



❶ Set up the apparatus as shown in the diagram, with the end of the glass tube under the water surface.

❷ Warm the air in the flask by placing your hands around the flask.

1 **a** What did you see in the beaker of water?
b What did this show was happening to the air in the flask?

❸ Watch as your teacher heats the flask with a Bunsen burner.

2 **a** What did you see in the beaker of water?
b What did this show was happening to the air in the flask?

❹ The flask will then be left to cool.

3 What did you see happening when the flask was allowed to cool?

4 Use the particle model to write a brief explanation of the expansion of air when it is heated.

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Convection in air

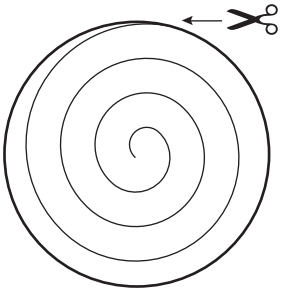
Aim

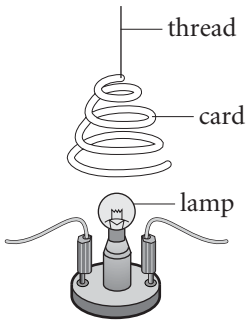
To investigate convection in air.

Equipment

- 12 V, 24 W lamp in holder
- connecting leads
- 12 V power supply
- thin card
- thread
- scissors
- pin

What to do





1. Draw a spiral shape on the piece of card, as shown in the diagram.
2. Cut out the spiral shape.
3. Use the pin to make a small hole in the top of your card shape.
4. Tie a length of thread through the hole.
5. Hold the card shape a few centimetres above the lamp and switch on the lamp.

Questions

1 Describe briefly what happens to the card spiral when it is held above the hot lamp.

2 Explain what makes the card move. Include these words in your explanation:

heated air		**cooler air**
less	**more**	**dense**

3 Explain why a hot air balloon rises.

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Convection in water

Aim

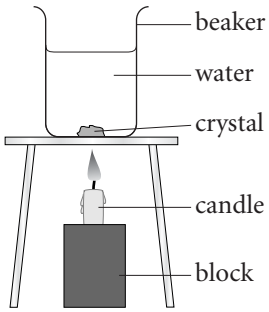
To investigate convection in water.

Equipment

- beaker
- drinking straw
- small crystal of potassium manganate(VII)
- tripod and gauze
- heatproof mat
- candle
- forceps

What to do

1. Pour water into the beaker until it is about three-quarters full.
2. Place the beaker on the tripod and gauze and allow it to stand until you are sure the water is not moving.
3. When the water is still, place your finger over one end of the straw and put the other end carefully into the water until it is touching the middle of the bottom of the beaker.
4. Take your finger off the top of the straw and carefully, using the forceps, drop the crystal of potassium manganate(VII) down the straw.
5. When the crystal has reached the bottom of the beaker, very carefully remove the straw, trying not to disturb the water.
6. Using the candle, gently heat the bottom of the beaker in the middle. Watch what happens to the crystal of potassium manganate(VII).
7. Make a large copy of the diagram opposite and draw on it the movement of the colour that comes from the crystal.



Questions

1 What did you see as you heated the water? What does this show you about what happens when water is heated?

2 Explain what causes hot liquids to rise, including the following words:

expansion (or expand)	**density (or dense)**

3 Use the particle model to explain the process of convection. You may wish to draw diagrams to help with your explanation.

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